A LOGARITHMIC BOUND FOR UNIONS OF PERMUTATION STRIPS

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Abstract. For the n parallelograms obtained by matching the equal subintervals of the two horizontal edges of the unit square, we prove that the minimum area of their union is of order $\frac{1}{\log n}$, as conjectured by Kaczynski.

For $n \geq 2$, divide the top and bottom edges of the unit square into intervals $I_i = [\frac{i-1}{n}, \frac{i}{n}]$, $1 \leq i \leq n$, and, given a permutation $\sigma \in S_n$, join bottom interval I_i to top interval $I_{\sigma(i)}$ by a parallelogram. We call these parallelograms *strips* and write α_n for the least possible area of their union as σ varies. Although their end intervals fill both edges completely, the strips may overlap between the edges, and we ask how small their union can be. More generally, a strip with bottom interval $[a, a+u]$ and top interval $[b, b+u]$ has horizontal section $[(1-t)a+tb, (1-t)a+tb+u]$ at height t, whose length u we call the width of the strip. The area of a union of strips is the integral of the length of its section over $0 \leq t \leq 1$.

The geometric question already appears in Kaczynski's thesis [1, Theorem 9 and Problem 5]: for a disjoint family of segments joining two parallel lines, the trapezoid formula remains valid with lengths and area interpreted as outer measures, and Problem 5 asks what lower bound can remain for a Borel union when the segments are allowed to cross. The permutation problem keeps both edges covered by intervals of equal length but allows their order to differ on the two edges.

Kuklinski [2, § 3, Conjecture 2] records Kaczynski's conjecture that α_n lies between positive constant multiples of $\frac{1}{\log n}$, and reports that Kaczynski claimed both the logarithmic lower bound and an upper bound with an additional factor of order $(\log\log n)^2$. Kuklinski's Theorem 4 gives $\alpha_n \leq \frac{C}{(\log n)^p}$ for some $p > 0$ through the projection estimates of Bond and Volberg [3]. We obtain the conjectured order through the following bounds.

Theorem 1. *There is an absolute constant C such that, for every $n \geq 2$,*

$$\frac{1}{H_n} \leq \alpha_n \leq \frac{C}{\log n}, \qquad H_n = \sum_{j=1}^{n} \frac{1}{j}.$$

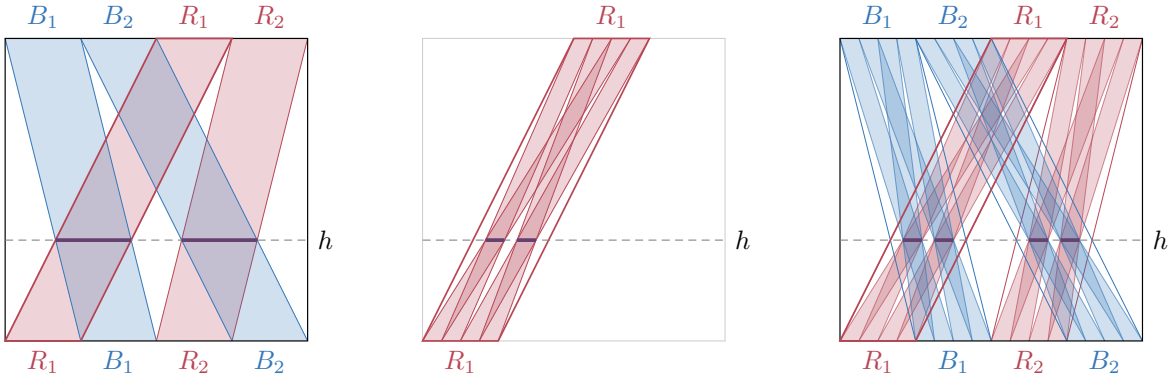
If height is regarded as time, the section formula describes intervals of constant length moving at constant speed. For the reversal $\sigma(i) = n+1-i$, their sections coincide at $t = \frac{1}{2}$, but the length of their union at height t is $\frac{1}{n} + \frac{n-1}{n}|1-2t|$, whose average is $\frac{n+1}{2n}$ and tends to $\frac{1}{2}$. We will instead place copies of an entire arrangement so that each copy retains the old intersections and paired sections coincide at a height chosen at each step. To obtain the logarithmic bound, we must compare how much area is lost with how many strips this requires.

1. The geometric construction

We first allow strips of unequal widths, with the same width at both ends of each strip, and require their end intervals to cover each edge with disjoint interiors. At the end we will replace these strips by strips of width $\frac{1}{n}$, with a bound on the area added.

Choose $0 < w \leq 1$ and divide $[0,1]$ into $m = \lceil \frac{1}{w} \rceil$ intervals, the first $m-1$ of length w and the last of length $1-(m-1)w$. Write R_i and B_i for red and blue copies of the ith interval. On both horizontal edges of $[0,2] \times [0,1]$, place these copies consecutively from left to right in the following orders, and join intervals with matching labels:

$$\begin{aligned} \text{bottom:} &\quad R_1, B_1, R_2, B_2, \ldots, R_m, B_m, \\ \text{top:} &\quad B_1, B_2, \ldots, B_m, R_1, R_2, \ldots, R_m. \end{aligned}$$



For $w = \frac{1}{2}$, the left panel shows the four enclosing strips after division of horizontal coordinates by 2. The middle panel isolates a copy of this arrangement inside R_1, while the right panel shows all four copies, giving sixteen strips. The marked sections lie at $h = \frac{1}{3}$. The repeated choice $w = \frac{1}{2}$ illustrates the copying rule. In the proof, w is chosen anew at each step.

Within each colour, the end intervals have disjoint interiors and occur in the same order on both edges. Multiplying the bottom endpoint inequalities by $1 - t$ and the top ones by t, then adding, shows that their sections have disjoint interiors at every height. For each pair of width w, the blue interval starts w to the right of the red one on the bottom edge and one unit to its left on the top edge. The difference between their left endpoints at height t, blue minus red, is therefore $(1 - t)w - t = w - (1 + w)t$. Since both sections have length w, they coincide at $h = \frac{w}{1+w}$. We can prescribe any $0 < h \le \frac{1}{2}$ by taking $w = \frac{h}{1-h}$. The shorter final pair, when present, need not have coincident sections at this height.

Now let N be the number of strips in the arrangement we wish to improve, reserving n for the prescribed final number, and let $E_t \subset [0, 1]$ be the union of its horizontal sections at height t. Writing $|E_t|$ for the length of this union, its area is $A = \int_0^1 |E_t|\, dt$.

For an enclosing strip with bottom interval $[a, a + u]$ and top interval $[b, b + u]$, replace each point (x, t) of the old arrangement by

$$(ux + (1 - t)a + tb,\ t).$$

At each height, this multiplies interval lengths and intersection lengths by u, and at the two edges it carries the old partitions onto the end intervals of the enclosing strip. Apply this rule to all $2m$ enclosing strips, then divide every horizontal coordinate by 2. The resulting $N' = 2mN$ strips lie in the unit square and their end intervals partition both edges.

Write R_t and B_t for the unions of the new red and blue sections, $F_t = R_t \cup B_t$, and $A' = \int_0^1 |F_t|\, dt$. The enclosing intervals of either colour have disjoint interiors and total length 1 before division by 2. Each copy contributes length $\frac{u}{2}|E_t|$, so adding these lengths and using the formula for the length of a union gives

$$|R_t| = |B_t| = \frac{|E_t|}{2}, \qquad |E_t| - |F_t| = |R_t \cap B_t|.$$

It follows that $\frac{1}{2}|E_t| \le |F_t| \le |E_t|$, and integration gives $\frac{A}{2} \le A' \le A$.

2. Choosing the height

The overlap identity below will give an expected decrease in area of at least a constant times A^2 when we average over h. Since $\frac{1}{A'} - \frac{1}{A} = \frac{A-A'}{AA'} \ge \frac{A-A'}{A^2}$, this provides a fixed positive expected increase in reciprocal area, while successive multiplications of the number of strips become additions after taking logarithms. The lemma compares these two increases for a single choice of h, in a form that we can sum over successive steps. All logarithms below are natural.

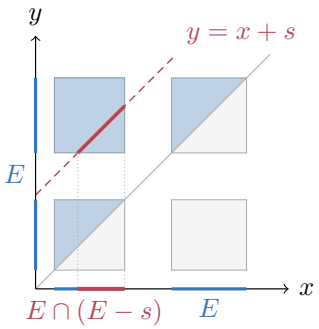
Lemma 1. *There is an absolute constant $K > 0$ such that any arrangement of N strips whose end intervals partition both edges, with union area A, admits a replacement of the same kind with $N' \geq 2N$ strips and area $\frac{A}{2} \leq A' < A$ satisfying*

$$\log \frac{N'}{N} \leq K \left(\frac{1}{A'} - \frac{1}{A} \right). \tag{1}$$

For a finite union $E \subset [0,1]$ of intervals, write $E - s = \{x - s : x \in E\}$ and let $1_E(x)$ be 1 for $x \in E$ and 0 otherwise. Writing the intersection length as an integral, changing the order of integration and putting $y = x + s$ gives

$$\int_0^1 |E \cap (E - s)| \, ds = \int_0^1 \int_x^1 1_E(x) 1_E(y) \, dy \, dx = \frac{|E|^2}{2}. \tag{2}$$

The double integral is the area of the part of $E \times E$ with $x \leq y$. Exchanging x and y interchanges its two halves, while the diagonal has area zero, proving the last equality. The integral in (2) depends only on $|E|$, so we can apply the identity to E_t without knowing the positions of its component intervals.[1]



The shaded region $\{(x,y) \in E \times E : x < y\}$ has area $\frac{|E|^2}{2}$. The part of $y = x + s$ inside it is marked in red, together with its set of x-coordinates $E \cap (E - s)$ on the horizontal axis. Equation (2) integrates this horizontal length over s.

Proof of Lemma 1. If necessary, replace every point (x,t) of the old arrangement by $(x, 1-t)$, interchanging its two edges without changing its area or number of strips. We may then assume $\int_0^{\frac{1}{2}} |E_t| \, dt \geq \frac{A}{2}$, where E_t now denotes the sections after this possible interchange. We keep this old arrangement fixed and vary h, which changes both the widths and positions of the new copies, and hence F_t, A' and N', while leaving E_t unchanged.

For a red and blue pair arising from an interval of length w, the left endpoint of the blue enclosing interval minus that of the red one is $\frac{w}{2}(1 - \frac{t}{h})$ at height t. Subtracting the red left endpoint and dividing horizontal coordinates by $\frac{w}{2}$ turns the two copied sections into E_t and $E_t + 1 - \frac{t}{h}$. Their intersection in the square therefore has length $\frac{w}{2}|E_t \cap (E_t + 1 - \frac{t}{h})|$. Since there are $\lfloor \frac{1}{w} \rfloor$ such pairs, their red enclosing strips have disjoint interiors, and $w\lfloor \frac{1}{w} \rfloor \geq \frac{1}{2}$, we may add these overlaps without overcounting to obtain

$$|E_t| - |F_t| \geq \frac{1}{4} \left| E_t \cap \left(E_t + 1 - \frac{t}{h} \right) \right|. \tag{3}$$

To apply (2), fix $0 < t \leq \frac{1}{2}$ and put $s = \frac{t}{h} - 1$. As h decreases from t to $\frac{t}{2}$, s increases from 0 to 1. Since $h = \frac{t}{1+s}$, changing variables gives $|dh| = \frac{t}{(1+s)^2} \, ds$. Averaging with constant density therefore leaves a factor t, whose reciprocal is not integrable near 0, preventing the Cauchy–Schwarz estimate below from giving a uniform bound. We instead average over one choice of h for the entire arrangement, with probability density $\frac{1}{\sqrt{2h}}$ on $(0, \frac{1}{2})$. The same h determines every section of the new arrangement, while in the estimate for each fixed t we retain only $h \in [\frac{t}{2}, t]$.[2]

[1] In convolution notation, $(1_{-E} * 1_E)(s) = \int_{\mathbb{R}} 1_E(x) 1_E(x+s) \, dx = |E \cap (E - s)|$. Tao [4] explains this overlap interpretation and remarks that in simple cases one can "work from first principles directly from the definition", as the substitution $y = x + s$ does here.

[2] This density belongs to a larger family: for any fixed $0 < a < 1$, the probability density $(1-a)2^{1-a}h^{-a}$ on $(0, \frac{1}{2})$ gives, after the same change of variables, a lower bound with weight t^{1-a}. Since $\int_0^{\frac{1}{2}} t^{a-1} \, dt$ and the expected value of $\log \frac{N'}{N}$ are both finite, the argument proceeds unchanged, with constants depending on a.

This gives a factor proportional to $\sqrt{t}$, whose reciprocal is integrable:

$$\frac{|dh|}{\sqrt{2h}} = \frac{\sqrt{t}}{\sqrt{2(1+s)^3}}\,ds \geq \frac{\sqrt{t}}{4}\,ds \qquad (0 \leq s \leq 1).$$

Writing $\mathbb{E}$ for expectation over this choice of h, we combine (3) and (2) and discard the nonnegative contribution from other values of h to obtain $\mathbb{E}[|E_t| - |F_t|] \geq \frac{\sqrt{t}}{32}|E_t|^2$.

To relate the integral of this reduction to the old area, apply Cauchy–Schwarz to $t^{\frac{1}{4}}|E_t|$ and $t^{-\frac{1}{4}}$ on $(0, \frac{1}{2})$, using $\int_0^{\frac{1}{2}} \frac{dt}{\sqrt{t}} = \sqrt{2}$ and the bound on the area below $t = \frac{1}{2}$ to get

$$\frac{A^2}{4} \leq \left(\int_0^{\frac{1}{2}} |E_t|\,dt\right)^2 \leq \sqrt{2}\int_0^{\frac{1}{2}} \sqrt{t}\,|E_t|^2\,dt.$$

Integrating this expected reduction over $0 < t \leq \frac{1}{2}$ therefore gives

$$\mathbb{E}(A - A') \geq \frac{1}{32}\int_0^{\frac{1}{2}} \sqrt{t}\,|E_t|^2\,dt \geq cA^2, \qquad c = \frac{1}{128\sqrt{2}}. \tag{4}$$

Although small values of h require more copies, the expected increase in $\log N$ remains bounded, allowing us to compare it with the increase in $\frac{1}{A}$. Since $2m \leq 2(\frac{1}{w} + 1) = \frac{2}{h}$ and $\frac{1}{A'} - \frac{1}{A} \geq \frac{A - A'}{A^2}$, the estimate (4) gives

$$\mathbb{E}\log\frac{N'}{N} \leq \mathbb{E}\log\frac{2}{h} = 2 + \log 4, \qquad \mathbb{E}\left(\frac{1}{A'} - \frac{1}{A}\right) \geq c.$$

Both expectations are finite, the first by the displayed bound and the second because $A' \geq \frac{A}{2}$. To compare the increases at the same height, we average their difference and choose K sufficiently large that

$$\mathbb{E}\left[K\left(\frac{1}{A'} - \frac{1}{A}\right) - \log\frac{N'}{N}\right] \geq Kc - (2 + \log 4) > 0.$$

The expression inside the expectation is therefore positive for some h, which gives (1) and also forces $A' < A$ because $N' \geq 2N$. □

3. Iteration and equal widths

We first construct an arrangement with $N < \sqrt{n}$, so that the replacement by equal-width strips below adds at most $\frac{2}{\sqrt{n}}$ to its area. For $\log n > 4K$, begin with the square, so $A = N = 1$, and apply Lemma 1 until the first stage at which $A \leq \frac{4K}{\log n}$. This stage is reached after finitely many steps because each application increases $\frac{1}{A}$ by at least $\frac{\log 2}{K}$. The preceding area exceeds $\frac{4K}{\log n}$, and a single step reduces it by at most half, so

$$\frac{2K}{\log n} < A \leq \frac{4K}{\log n}.$$

The upper bound gives the desired area, while the lower bound controls the number of strips: summing (1) cancels the intermediate values of $\log N$ and $\frac{1}{A}$, giving

$$\log N \leq K\left(\frac{1}{A} - 1\right) < \frac{1}{2}\log n,$$

so $N < \sqrt{n}$.

To obtain exactly n equal-width strips, use the original intervals $I_1, \ldots, I_n$ on both edges. Within each old strip, pair whole such intervals at the bottom and top one to one until one of the two ends has none left. An end interval of length u contains at least $nu - 2$ whole intervals: omitting the two possibly incomplete pieces at its ends removes less than $\frac{2}{n}$ of its length. Since both ends of an old strip have the same length u, at least $nu - 2$ pairs can therefore be formed inside it. Each strip formed by a matched pair stays inside the old strip, because its left and right endpoints lie between the old endpoints on both edges, and these inequalities remain true after taking their weighted averages with coefficients $1 - t$ and t. As the old widths sum to 1, at least $n - 2N$ intervals are matched on each edge. The numbers of unused intervals are equal and at most $2N$, so choose any

one-to-one pairing between them. Completing the pairing gives a permutation of all n intervals, and each additional strip has area $\frac{1}{n}$, giving final union area at most $A + \frac{2N}{n}$. Hence

$$\alpha_n \le \frac{4K}{\log n} + \frac{2}{\sqrt{n}} \le \frac{C}{\log n}.$$

Here C is an absolute constant. For the remaining n, use $\alpha_n \le 1$ and increase the constant.

4. The lower bound

For the lower bound, we estimate pairwise overlap from the endpoint separations, following the argument for the elementary Kakeya bound in Babichenko et al. [7, Proposition 6.5]. Let $Z(x,t)$ count the strips containing (x,t) and let A be their union area. Integrals without limits in this section are area integrals over the unit square. Then $\int Z = 1$ because the strip areas add to 1, and Cauchy–Schwarz on the union yields

$$1 = \left(\int Z\right)^2 \le A \int Z^2.$$

Expanding Z^2 and integrating gives the sum of the strip areas plus twice the sum of the pairwise intersection areas.

If two strips have the same order on both edges, their left endpoints are separated by at least $\frac{1}{n}$ at both ends and hence at every height, so their interiors do not intersect. For a pair with $i < j$ and $\sigma(i) > \sigma(j)$, put $p = j - i$ and $q = \sigma(i) - \sigma(j)$. The left endpoint of the second section minus that of the first is $\frac{p-(p+q)t}{n}$, so their intersection has length $\frac{1}{n}\max\{0, 1 - |p - (p+q)t|\}$. Since $p, q \ge 1$, substituting $s = p - (p+q)t$ in the integral over $0 \le t \le 1$ gives intersection area

$$\frac{1}{n(p+q)} \int_{-1}^{1} (1 - |s|)\, ds = \frac{1}{n(p+q)} \le \frac{1}{4n}\left(\frac{1}{p} + \frac{1}{q}\right).$$

For pairs that do not cross, use $p = j - i$ and $q = |\sigma(j) - \sigma(i)|$. The same upper bound still holds, since their intersection area is zero. On either edge there are exactly $n - d$ pairs whose indices differ by d, so summing over all pairs gives

$$\int Z^2 \le 1 + \frac{1}{n} \sum_{d=1}^{n-1} \frac{n-d}{d} = H_n.$$

Thus $A \ge \frac{1}{H_n}$, completing the proof of Theorem 1. □

In $\frac{1}{n}\sum_{d=1}^{n-1} \frac{n-d}{d}$, the terms with $2^k \le d < 2^{k+1}$ contribute at most 1: there are at most 2^k terms, each at most $\frac{1}{2^k}$. There are at most $1 + \frac{\log n}{\log 2}$ nonempty ranges, which explains the logarithm. The construction attains this order while preserving the partitions of both edges.

5. Remarks

There is also a useful description in terms of pairs of endpoints. For each strip joining intervals I and J, form the square $I \times J$, and let Q be the union of these squares. The sets of first and second coordinates of Q are both $[0,1]$, while

$$E_t = \{(1-t)a + tb : (a,b) \in Q\}.$$

The copying rule replaces each point (x,y) of Q by $(a + ux, b + uy)$ for an enclosing strip with end intervals $[a, a+u]$ and $[b, b+u]$. The final division of horizontal coordinates by 2 then divides both coordinates by 2. Before this division, a paired blue copy is obtained from its red copy by adding w to the first coordinate and subtracting 1 from the second, provided the original interval has length w. Under the formula for E_t, this changes each value by $(1-t)w - t$, which is zero at $t = h$. The paired copies therefore give identical sections at that height.

This relates the construction to Kuklinski's use of projection lengths [2, § 4] and to the Favard length studied by Bond and Volberg [3, § 1], which averages orthogonal projection lengths over all directions. Explicitly, let $L_Q(\theta)$ be the length of $\{a\cos\theta + b\sin\theta : (a,b) \in Q\}$. The substitution $t = \frac{\sin\theta}{\cos\theta+\sin\theta}$ gives

$$A = \int_0^{\pi/2} \frac{L_Q(\theta)}{(\cos\theta + \sin\theta)^3}\, d\theta.$$

Our area integral thus uses only the directions between the coordinate axes, with the weight displayed above.

The use of translated copies to reduce union area also occurs in the classical Perron–Schoenberg construction described by Besicovitch [5, pp. 700–705] and the quantitative version given by Keich [6, § 2]. Here the specified red and blue placements make paired sections coincide at a chosen height while preserving the partitions of the two edges.

The two estimates also show that the logarithmic order persists when heights are sampled with any fixed bounded probability density: if $0 \le \rho \le M$, $\int_0^1 \rho(t)\,dt = 1$, and E_t^σ denotes the section of the arrangement given by σ, then

$$\frac{1}{MH_n} \le \min_{\sigma\in S_n} \int_0^1 \rho(t)|E_t^\sigma|\,dt \le \frac{MC}{\log n}.$$

The upper bound follows from the same permutations as Theorem 1, while the lower bound follows by applying Cauchy–Schwarz with the measure $\rho(t)\,dx\,dt$, since the strip widths add to 1 at every height and therefore $\int \rho Z = 1$, whereas $\int \rho Z^2 \le M \int Z^2 \le MH_n$.

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References

[1] T. J. Kaczynski, *Boundary Functions*, PhD thesis, University of Michigan, 1967.

[2] P. Kuklinski, *An uncountable union of line segments with null two-dimensional measure*, arXiv:2311.16210 [math.CA], 2023.

[3] M. Bond and A. Volberg, *Buffon needle lands in ϵ-neighborhood of a 1-dimensional Sierpinski Gasket with probability at most $|\log\epsilon|^{-c}$*, C. R. Math. Acad. Sci. Paris **348** (2010), 653–656.

[4] T. Tao, *Computing convolutions of measures*, *What's new*, 26 July 2013.

[5] A. S. Besicovitch, *The Kakeya problem*, Amer. Math. Monthly **70** (1963), 697–706.

[6] U. Keich, *On L^p bounds for Kakeya maximal functions and the Minkowski dimension in $\mathbb{R}^2$*, Bull. London Math. Soc. **31** (1999), 213–221.

[7] Y. Babichenko, Y. Peres, R. Peretz, P. Sousi and P. Winkler, *Hunter, Cauchy rabbit, and optimal Kakeya sets*, Trans. Amer. Math. Soc. **366** (2014), 5567–5586.